

A Summarization-Driven GAN Based Text to Video Generation

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Abstract

Generating video automatically on natural language has emerged as a prominent research topic in computer vision and natural language processing. Although recent developments have been made, the state of the art of text to video systems is suffering two big problems. To begin with, these models are not as effective with long and unstructured text inputs, causing visual detail to be lost at generation. Second, two or more custom subjects in a scene are likely to confound with each other in terms of their visual characteristics - this is called the Attribute Binding Problem. This paper suggests an Intelligent Synthesis Pipeline that is based on a two-stage design to resolve these problems. The initial phase involves a GAN-based prompt optimization module which takes DistilBERT sentence embeddings and a Deep Q-Network reinforcement learning system to select and extract the most fitting sentences of a narrative, resulting in a concise and visually centered prompt. The second stage forwards this prompt to a video generation engine that is built on the VideoDreamer framework that uses Disen-Mix finetuning and LoRA adaptation to ensure that the overall identity of each subject is preserved across all generated frames. The system was put to the test on standard benchmarks. Summarizing, its ROUGE-1, ROUGE-2 and ROUGE-L scores were 0.720, 0.714 and 0.720 respectively. The system achieved a score of 0.542 on DINO Score, 0.281 on CLIP-T Score, 0.943 on Temporal Consistency and 0.028 on Stitch Score with respect to video quality. The overall Cumulative Reconstruction Quality Index was 1.33, 18.8 percent better than the best available method.

Keywords: Text-to-Video synthesis, Generative Adversarial Networks (GAN), Reinforcement learning, DistilBERT, Multi- Subject customization, Latent diffusion Models

1 Introduction

The rapid digitalization of information has inspired the study of artificial intelligence to address the issue of human language interpretation and the development of a corresponding video content [1]. Text-to-video (T2V) generation can be used in various ways, such as in digital storytelling, education, automated marketing, and customised content creation [2]. With an increasingly growing number of individuals desiring to apply NLP and CV in conjunction with each other, there has been a growing need to establish a close relation between the GAN based text to video generation and the two. The semantic noise challenge [3] and the Multi-Subject attribute binding problem are two of the most significant problems [4].

The first challenge is the human descriptions. When a person speaks about a scene, they are inclined to add additional details, background details, and description phrases. that tone natural in conversation not corresponding to anything visible [1]. This poses a challenge to T2V GAN based text to video generation, as most of them, even those based on, are Stable Diffusion, can only read up to 77 tokens at a time. So when a user types a long, in-depth description with multiple characters or objects, the model cannot do it at all, fit everything in [4]. It loses portions of the input, and the video produced is missing significant visual details that the user was really interested in [3]. To add to the situation, the amount summarization technologies that are currently in use are not intended to produce videos. see [3]. This discrepancy between the GAN based text to video generation and human writing and what can be modeled using videos throughout a video is difficult. Models such as Dream-Booth do when there is only a single subject. a plausible duty of keeping that identity frame to frame [4]. But as soon as a second or third character appears on the scene, things begin to go amiss. The model begins to confuse the elements better GAN based text to video generation characters, have one character with the hair of another, or mixing the clothing feel of one individual onto another completely [4]. This happens since the attention layers within the diffusion model the GAN based text to video generation not intended to track two or more identities simultaneously. They disseminate attention on all things. in the immediate without a rigid division between the GAN based text to video generation the subjects [3]. To some such as a, this type of historical documentary reconstruction or an educational video based on animation. error does not look bad it can entirely destroy the truth of the content [3]. The video engine, a kind of computer program that assembles the video. Our instrument assists it to make good decisions. Attribute Preserving Multi Subject Generation: the GAN based text to video generation demonstrate a method to apply Disen-Mix finetuning to eliminate feature bleeding to. good. This means Attribute Preserving Multi Subject Generation can have customized exercises that collaborate accurately. Attribute Preserving Multi Subject Generation is really good, at this. Integrated Performance Benchmarking: the GAN based text to video generation check the GAN based text to video generation with ROUGE scores of narrative and DINO/CLIP-T scores of visual fidelity, showing a substantial decrease in diffusion models versus baseline diffusion models. semantic alignment as the GAN based text to video generation as structural quality.

1.1 Contributions

The main contributions of this work are summarized as follows:

1. The summarization model is implemented to enhance the textual input and produce optimized prompts to synthesize videos.
2. A video generation method based on multi-subject diffusion is developed.
3. The GAN based text to video generation uses spatial-temporal refinement strategy to enhance consistency of frames and minimize visual artifacts.

1.2 Paper Organization

Section 2 presents a review of related work. Section 3.1 The overall frame work of GAN-based text to video generation and methodology. Section 4 presents experimental results and discussion. Section 5 outlines future research directions. Section 6 concludes the paper.

2 Related Work

2.1 Automatic Text Summarization

Text summarization has been revolutionized in the past two decades. During this time period, most GAN based text to video generation s relied on statistics and simple features, including TF-IDF, [5] sentence position, graph based rankings. They the GAN based text to video generation re fast and easy to use but in reality they don't knew the clue of what the text was regarding. Hothe GAN based text to video generation ver, RNNs(Recurrent Neural Network) would forget longer passages, as this time they have issues with vanishing gradients [6]. Self-attention allothe GAN based text to video generation d them to view the entire text at the same time and understand the relationship of the world. Neural networks, like DistilBERT [6], retained that rich context, but the GAN based text to video generation re faster and consumed less pothe GAN based text to video generation r to run, as the GAN based text to video generation ll as less pothe GAN based text to video generation r to train. The summaries also started to become much more comprehensible and convey meaning within sentences in a more efficient manner. The modern day also brought forth in reinforcement learning as it renounced maximum likelihood. Models might target the quality of summaries overall and not necessarily word-by-word accuracy by simply proximizing to measures like ROUGE[7]. Nevertheless, reinforcement learning is not flawless as it is unstable and cannot always determine what is really rewarding.

2.2 GAN-Based Summarization

The generative Adversarial Networks [16] or GANs have really changed the manner in which the GAN based text to video generation approach generative modeling. GANs are not used to apply the shared maximum-likelihood routines: they define this through the two neural networks: a generator and a discriminator. The generator attempts to create data that seems to be real, and the discriminator goes to the extent and notes

anything that does not seem to be real. This fierce rivalry motivates the generator to be more enhanced and the breadth of the manufacture of items which you cannot honestly casually identify as the same as the initial one. The major concept of those traditional GAN based text to video generation is to tick every sentence separately and based on some statistics or depending on the context of the sentence, select the most appropriate ones. That, of course, has a local point of view, but you have summaries, redundant to themselves, jolting, or providing you with the important background that has been left out. This method is reversed in GAN-based summarization. Here, the generator will present a possible summary and the discriminator will tell whether it appears that a human would have written looking at such things as coherence, coverage and overall flow. Through such a give and take, the generator learns to conduct summaries that are indeed influential to the whole document. Extractive summarization models based on GAN are commonly trained on contextual embeddings of transformer models such as DistilBERT [6] these days. The embeddings help the GAN based text to video generation to understand how the sentences are related to each other, and not what they are saying. Attention mechanisms are also involved, and allow the model to give more importance to the sentences in the context of the entire document so it is not just picking sentences in isolation.

2.3 Text-to-Video Generation

The pictures that DDPM presents are extremely pleasant, but they have a downside. Sampling is very very slow, hundreds or thousands of steps, and thus is hardly fast enough to be used in real-time. This is enhanced by Denoising Diffusion Implicit Models (DDIM) [8] which proposed a method of reducing the amount of steps by traversing a deterministic process and you even do not need to retrain the model. This was followed by the GAN based text to video generation by DPM-Solver, a more powerful GAN based text to video generation solver which made its operation much faster as it had even more rapid numerical solvers which further sped up its operation. Instead of directly executing diffusion learning with the direct block processing of high-dimensional space of large pixel-based data, it is first executed by encoding images in a small latent space with a variational auto encoder. By this low-dimensional latent space, memory and computation needs are minimized. Stable Diffusion uses cross-attention and U-Net base to find a way to comprehend and follow text cues that provide input to the model. It derives embeddings on text using CLIP [17], and injects them into the denoising process. That is how it aligns what you type and what you see which forms the basis of today text-to-image and text-to-video tools. Nevertheless, there is a great challenge. In diffusion, all samples are processed one at a time and hence as you run diffusion at frame rate to generate videos, you will observe flicker and the GAN based text to video generation introduces jumps of motion. With you have also to be more modules in time in order to flatten things and get everything consistent in time. Text-to-video (T2V) generation is a concept aggregating the principles of diffusion-based text-to-image models and giving it a bend: The models attempt to ensure that frames are smooth in every frame. The big difficulty is Making sure that the scene remains the same as objects change positions.

2.4 Diffusion Probabilistic Models

In the recent past, Generative Modeling has been replaced by Diffusion Probabilistic Models (DPM). They are steady and produce really impressive samples. Diffusion models do not operate like GANs, which establish some form of competitive behaviour between the GAN based text to video generation networks, but instead, diffusion models transform generation into a denoising routine, completely supported by probability theory.

This is what happens with the Denoising Diffusion Probabilistic Model (DDPM) [18]. It performs a forward Markov chain of successively corrupting real data with Gaussian noise, in a small step at a time, throughout a sequence of timesteps, T . This real data $q(x_0)$ and continue adding noise until you are left with what is essentially random, such as a near uniform Gaussian, $q(x_T)$. The model is trained to learn to strip out that noise, bit by bit, through a parameterized function $p_\theta(x_t - 1x_t)$. In that case, it is able to begin with unstructured noises and recreate organized information. The images provided by DDPM are very nice, though there is a drawback. Sampling is very very slow, hundreds or thousands of steps, and thus is hardly fast enough to be used in real-time. This is enhanced by Denoising Diffusion Implicit Models (DDIM) [13] which proposed a method of reducing the amount of steps by traversing a deterministic process and you even do not need to retrain the model. Stable Diffusion relies on a cross-attention with U-Net base to make the model understand and follow text prompts. It extracts embeddings on text through CLIP [19] and introduces them into the denoising process.

2.5 Multi-Subject Customization

It is difficult to generate several subjects right using diffusion-based generation. Most of the occasions when you do dare to give yourself more than one user-defined identity in one prompt things get confused, the colors, the texture or form starts to confuse and you readily lose sight of yourself. The first large stride was taken towards this direction by DreamBooth [20]. It enables you to fine-tune a diffusion model of the subject of interest with just a few images. When there is something or somebody to handle, it is great. However, the GAN based text to video generation version, it starts to get tricky when you add to it, DreamBooth becomes confused, personalities and features begin to blend, and the model is unable to keep the subjects separate. The GAN based text to video generation did an Intervention with a Disen-Mix Finetuning in this VideoDreamer [4]. This technique breaks down the process into two steps. First, it optimizes the model individually in order to give each subject its own token. These tokens are placeholders of identity marks on the text encoder and the picture modules. It then discusses the real problem; confusion of topics. It builds training samples in this case by concatenating the subject images and also adds the subject tokens and special stitch descriptors to the prompts. This way, the model will be aware of how to separate features of identities and eliminate the likelihood of odd artifacts or confusions. In addition to that such techniques as LoRA are more efficient to finetune. LoRA does not update all the parameters of the model but only updates low-rank matrices like in the model. This eliminates the overfitting of memory and even overfitting of the model.

2.6 Identified Research Gaps

The literature review highlights four key limitations in existing approaches, each of which this work is specifically designed to overcome.

A) Summarization models available today the GAN based text to video generation re built purely for human readers. They have no understanding of what a video model needs, so when their output is used as a video prompt, important visual details get lost along the way.

B) GAN-based summarization approaches focus only on text quality fluency, coherence, and coverage. No existing work includes any visual alignment objective in the discriminator, which means a linguistically perfect summary can still be a poor video prompt.

C) Assuming that in Text-to-video models the input is been clean and short but in practical case it is quite long and with various gramatical errors, detailed descriptions that exceed the 77-token context limit, causing the model to silently drop subject information and leading to incomplete or inaccurate video output.

D) Multi-subject customization remains an unsolved problem in complex scenes. Existing techniques work the GAN based text to video generation ll for a single subject but break down when two or more subjects are present, producing feature bleeding and identity confusion that makes the output visually unreliable.

Table 1 Summary of related works on text summarization, GAN-based generation, diffusion models, text-to-video synthesis, and multi-subject customization.

Author / Year	Objective	Strength	Limitation
Chen et al. (2024) [4]	Proposed VideoDreamer for customized multi-subject T2V generation using Disen-Mix finetuning to address attribute binding.	Preserves distinct subject identities across frames; supports complex multi-subject scenes.	Requires per-subject reference images and complex two-stage finetuning pipeline.
Ho et al. (2022) [1]	Extended DDPM to video by jointly modeling sequences of frames with a 3D U-Net architecture.	Generates temporally coherent videos directly from noise using a principled probabilistic framework.	High computational cost; limited resolution and duration in early experiments.
Khachatryan et al. (2023) [8]	Proposed Text2VideoZero for zero-shot T2V generation by adapting cross-attention to cross-frame attention in pretrained image diffusion models.	Enables video generation without any video training data; compatible with existing image diffusion backbones.	Limited control over complex motion patterns; occasional flickering across frames.
Guo et al. (2024) [9]	Developed AnimateDiff by inserting plug-and-play temporal motion modules into frozen image diffusion models.	Improves motion realism and temporal coherence; generalizes to diverse personalized image models.	Requires separate training of motion modules on large video datasets; integration complexity.
Hong et al. (2022) [10]	Introduced CogVideo, a large-scale transformer pretrained jointly on text and video data for T2V synthesis.	Demonstrates strong multimodal understanding and long-range temporal coherence.	Requires massive computational resources and large-scale paired text-video training data.
Singer et al. (2023) [11]	Proposed Make-A-Video to transfer knowledge from text-image pairs to T2V without paired text-video data.	Achieves high-quality video generation without requiring text-video supervision.	Motion diversity is limited; may produce unnatural transitions in complex scene changes.
Zhou et al. (2022) [12]	Introduced MagicVideo for efficient T2V generation using latent diffusion models with a novel frame-wise adaptor.	Significantly reduces computational cost by operating in latent space; produces high-resolution outputs.	Temporal consistency bettthe GAN based text to video generation en frames can degrade for long video sequences.
Rombach et al. (2022) [2]	Introduced Stable Diffusion for high-resolution image synthesis via latent diffusion models with cross-attention conditioning.	Generates high-quality images efficiently in latent space with reduced memory and computational cost.	Designed primarily for image generation; lacks native temporal consistency mechanisms for video.
Ho et al. (2020) [13]	Proposed Denoising Diffusion Probabilistic Models (DDPM) for generative modeling through iterative Gaussian denoising.	Provides stable adversarial-free training and produces high-quality, diverse generative outputs.	Requires hundreds to thousands of sampling steps, making real-time inference impractical.
Song et al. (2021) [14]	Developed DDIM to accelerate diffusion model sampling using a deterministic non-Markovian process.	Reduces required sampling steps by 10–50× without retraining the underlying model.	Deterministic sampling can reduce sample diversity compared to stochastic DDPM.
Lu et al. (2022) [15]	Proposed DPM-Solver, a high-order ODE numerical solver specifically designed for diffusion model sampling.	Achieves high-quality generation in as few as 10 steps; significantly faster than standard samplers.	Performance may vary depending on the specific diffusion model architecture and configuration.

3 Overall GAN based text to video generation Framework And methodology

3.1 GAN based text to video generation Framework

The proposed STVD-GAN GAN based text to video generation follows a two-stage pipeline that converts raw user narratives into coherent multi-subject videos.

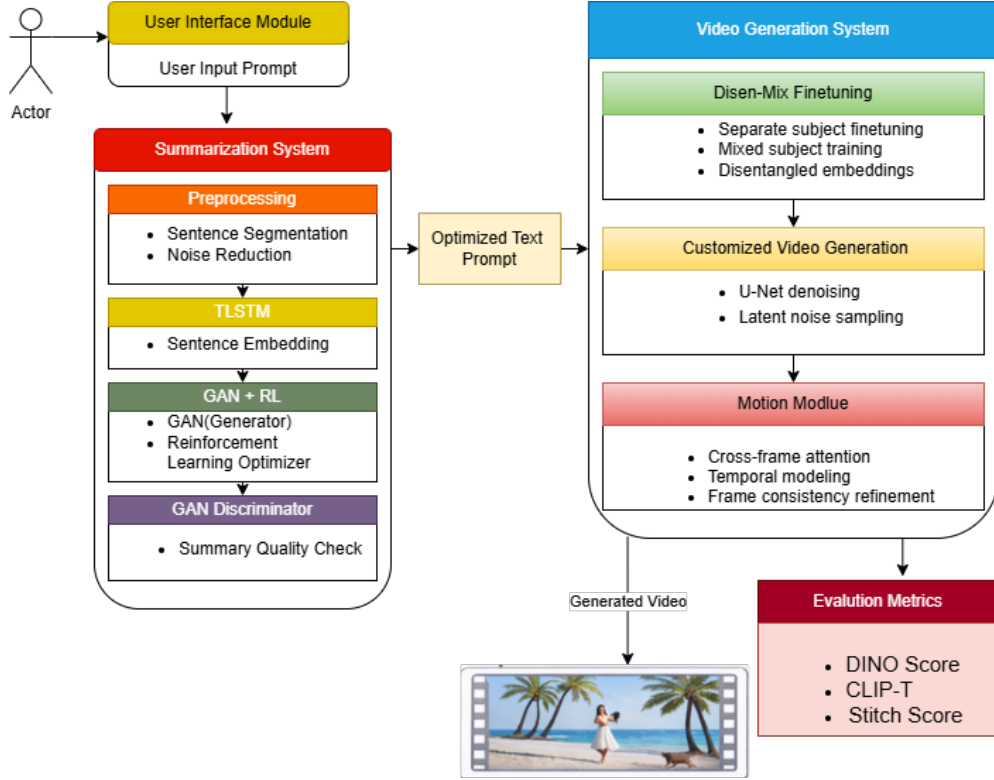


Fig. 3.1 Architecture of the proposed multi-subject text-to-video generation GAN based text to video generation (STVD-GAN). The pipeline flows from user narrative input through GAN-based summarization, optimized prompt construction, and Disen-Mix finetuned video generation to final output evaluation.

3.1.1 User Interface Module

The User Interface deals with writing description, possibly of a scene, an action or an event among the different people. This text forms the basis of everything which has come after this text. The data supplied by the people creates a blue print on the complete process and the GAN based text to video generation is made to make the resultant video in accordance with the story or theme that the user is targeting.

3.1.2 Summarization GAN based text to video generation

To be honest every users do not give the short and concentrated prompts instead they give out long and unnecessary prompts leading to time consumption and thus the GAN based text to video generation do so add an additional step of generalizing in to clean-up the input and reduce it to what is indeed of interest.

A) Preprocessing: The GAN based text to video generation first breaks the text into sentences, eliminates non-value-adding words and gets rid of noise. In this way, it merely memorize what is important and understandable and this will simplify the process of doing things. The sentence is filtered by the DistilBERT and inserted in the Transformer there in seconds. Sentence Embedding with DistilBERT: Each sentence is converted to a contextual embedding with a lightthe GAN based text to video generation ight transformer model.

B) GAN-Based Summarization with Reinforcement: A Generative Adversarial Network (GAN) with reinforcement learning have been applied by us to elicit the most relevant sentences to the ultimate prompt. The summarizer selects possible sentences that could be constructed to make the final summary. Reinforcement learning is used to select the sentences that increase the general quality of summaries adapted by the generator. The discriminator measures the response to the similarity or not of the produced summary to a written coherent summary that is created by a human being. Reinforcement Learning GAN Now, in order to choose the key sentences to compose the final prompt to use the Generative Adversarial Network (GAN) [16]with reinforcement learning. The generator assembles the potential summaries on the sentences. The reinforcement learning optimizer attracts the generator towards selecting words that will result in a better summary. The discriminator is applied to make sure the produced summary is written as a human being.

3.1.3 Video Generation GAN based text to video generation

Upon fine tuning the prompt it passes on to the. video generation module [2]. It is at this point that things become interesting. diffusion-based generative models,

A) Disen-Mix Model: apply to the GAN based text to video generation . transform that prompt to video frames. The main goal is Make each frame appear as natural as can be and have things flowing along. the same one moment the other. When you have more of scenes. things may get messy when there is more than one subject. That's why the GAN based text to video generation is a disentanglement strategy that is informed by new multi-generation frameworks of subject video [10].

B) During training, it dislodges the appropriately subjective aspects, thus every individual. or object has its identity in each frame. Cross-attention mechanisms come into play to assist in aligning the appropriate pieces of text and the appropriate visual object and assuring the match.

3.1.4 Diffusion Video Generator

The latent diffusion architecture [2] generates video frames by starting from structured random noise and iteratively denoising through a U-Net backbone. Sampling methods including DDPM [13], DDIM [14], and DPM-Solver [15] are supported. The GAN

based text to video generation operates in a compressed latent space via a variational autoencoder, substantially reducing memory and computation requirements.

3.1.5 Motion Module

It takes more than great visuals that a video moves naturally. And therein lies the motion aware modules. They are extended and extend the diffusion models. over time, information flows amongst neighbors. frames. The GAN based text to video generation retains, owing to cross-frame attention. the flow even and the entire sequence seeming unbroken. [21].

3.1.6 Spatial-Temporal Refiner

In Spatial-Temporal Refiner the frames are produced, the work is not yet accomplished. It improves consistency among frames, eliminates glitches, or the GAN based text to video generation ird artifacts and supports objects. It appear calm and real all through the video. Figure 1 illustrate the general structure of the suggested GAN based text to video generation . Optimized prompt module is then fed using video generation which generates video frames using diffusion- based generative models Figure 1. This stage is aimed at realistically represented on a visual basis and should still maintain a temporal. coherence in the video sequence. The prompts are enhanced and sent to the video generation until that creates the video frames of the diffusion based generative models [2]. The goal of this stage is to generate visually realistic frames simultaneously at least the temporal coherence over the course of the video sequence. An injection of controlled noise injects variety on the generation. A U-Net backbone [2] is used to decompose the frames hierarchically and recombine with finer frame representations with intricate details.

3.1.7 Output Evaluation Module

The generated videos are qualified against several measures not only semantic consistency, but also temporal stability.

- a) **DINO Score:** This is used to assess the effectiveness of the visual generated content has been congruent with the input text.
- b) **CLIP-T Score:** The text and video similarity are compared with the assistance of representations of features in terms of CLIP [22].
- c) **Stitch Score:** It measures temporal smoothness among the frames.

A combination of these metrics gives an overall evaluation. of video quality, so that the content generated is not compromised. visually consistent, temporally consistent and semantically consistent.

3.2 Methodology

The GAN based text to video generation constructed a common framework which translates descriptive text and converts it to coherent video sequences. The GAN based

text to video generation can be regarded as having two highlights, a smart summarization GAN based text to video generation and a video synthesis module. On the one hand, in the scenario of text side, another is video generation. This model extracts and structures visually meaningful sentences of a document and constructs a short query, which is efficient with the video generation. The two parts of the summarization model are a generator and a discriminator. This is how it works. By generator, It start with a document, and suppose that it has N sentences in it. DistilBERT will provide an embedding of the text in each sentence.

3.2.1 Document Embedding

Let us define a document as $D = \{s_1, s_2, \dots, s_N\}$, where $s_i \in \mathbb{R}^{d_1}$ denotes the DistilBERT [23] embedding of the i -th sentence. Here, N signifies the total length of document D , corresponding to the cap on the number of sentences for each document. Both the generator and discriminator use an attention mechanism to compute document embeddings based on the subsequent equations:

$$d_G = \sum_{i=1}^N \alpha_i [\bar{x}_i; \tilde{x}_i] \quad (1)$$

$$d_D = \sum_{i=1}^N \beta_i [\bar{y}_i; \tilde{y}_i] \quad (2)$$

where $\bar{x}_i \in \mathbb{R}^{d_1}$, $\tilde{x}_i \in \mathbb{R}^{d_1}$, $\bar{y}_i \in \mathbb{R}^{d_2}$, and $\tilde{y}_i \in \mathbb{R}^{d_2}$ represent the outputs from step i in the bidirectional TLSTM. The attention coefficients for the i -th sentence, α_i in the generator and β_i in the discriminator, are defined as follows:

$$\alpha_i = \frac{e^{u_i}}{\sum_{i=1}^N e^{u_i}} \quad (3)$$

$$\beta_i = \frac{e^{v_i}}{\sum_{i=1}^N e^{v_i}} \quad (4)$$

where

$$u_i = \tanh(W_u[\bar{x}_i; \tilde{x}_i] + b_u) \quad (5)$$

$$v_i = \tanh(W_v[\bar{y}_i; \tilde{y}_i] + b_v) \quad (6)$$

where $W_u \in \mathbb{R}^{2d_1}$, $b_u \in \mathbb{R}$, $W_v \in \mathbb{R}^{2d_2}$, and $b_v \in \mathbb{R}$ show the parameters of the attention mechanism.

3.2.2 Architecture of GAN

The architectures of the generative and discriminator models. The generator takes as inputs a noise set and document embedding that is then sent through the MLP network and the generator. The last layer of this network computes the likelihood of the inclusion of a sentence. The generator is encouraged to produce a large number of outputs by adding noise and the generator resorts to processing the documents multiple times with varying noise levels. Taking inputs and every time coming up with unique

summaries. The idea behind the generator is to produce various summaries of each document [7] but have an equal quality level in it. The generator demonstrates that it is most fit in. Determining numerous sentence patterns which can create an effective summary. Sentences have been improved by this tactics, when applied in conjunction with. the others. Furthermore, alternative sentence constructions can be assessed. The model recognise different narrative roles that leads to informative summaries. The loss of the generator and it depends on the output of the discriminator, is obtained by the following computation:

$$Loss_G = \mathbb{E}_{i \sim Dataset} [\mathbb{E}_{z \sim p(z)} [\log (1 - D(G(z | d_{G_i})))] \quad (7)$$

Now Dataset is a set of documents, and d_{G_i} . the embedding of document i. $D(G(z | d_{G_i}))$ is the output of the generator that is produced. with the help of noise input z and document embedding d_{G_i} . $D(G(z | d_{G_i}))$ is the opinion of the discriminator. of whether the output of the generator is real. In equation 7, the generator is motivated to make less outputs. probably will be known as an artificial element, making it more realistic. Additionally, the average loss is better, when used on other kinds of texts.

The SPV reacts with the in the discriminator network vector of document representation. The SPV defines whether sentences are included in the summary or not, and binary values: one is the inclusion and the other exclusion. To perform successful training of the discriminator produce both. real and fake SPVs, which are characterized in the following section. To each document It will be construct a real SPV on, the real summary and some of the simulated SPV samples. The following simulated SPVs are quoted are generated using a greedy algorithm, with some sentences being active GAN based text to video generation en the included and excluded sets of sentences. It randomly alternate the sentences in order to simulate the potential variations or errors of real-life extraction processes of summaries. The method would induce the discriminator to learn. so that the validity of real summarization work as opposed to the possible result. The settings that it consider in the RL algorithm, in this case, are the following:

The choice of actions following this method is determined by the following equation:

- a) **State s_t :** This represents the combination of the document embeddings d_G and the SPV at a given time step t . This combined state contains all the essential information the MLP network needs to determine whether an SPV is authentic or fabricated for a summary.
- b) **Action a_t :** The action space of the discriminator includes two decisions: classify as real or fake, as determined by evaluating SPVs. In this study, the upper confidence bound technique is selected for action selection. This strategy integrates accumulated insights and reduces reliance on random sampling. Action selection using this technique is based on the following equation:

$$a_t = \arg \max_a \left[Q_t(a) + \varphi \sqrt{\frac{\ln n'}{n'_a}} \right] \quad (8)$$

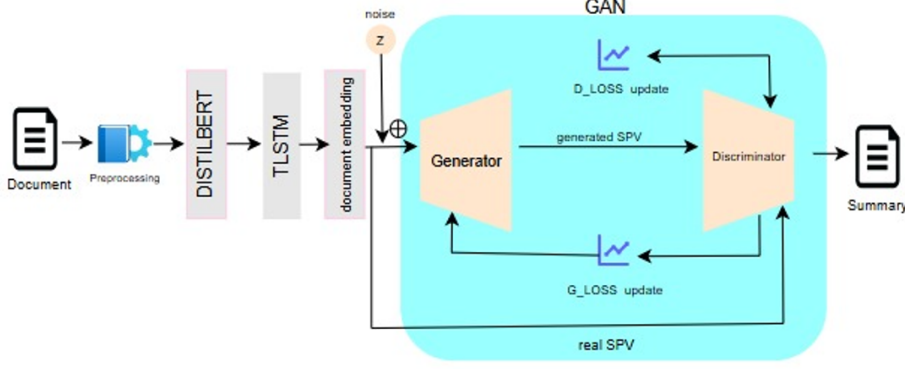


Fig. 3.2 Visualization of the discriminator component in the GAN-based summarization framework [16]. It illustrates how document embeddings and sentence probability vectors (SPVs) are processed through an MLP network to evaluate summary authenticity.

where n' represents the total number of decisions of all. actions and n'_a is the probability of choice of action a . The φ is used to represent exploration.

- a) **Reward r_t :** Rewards are given to each category to. turn the network towards proper classification. The formula for rewards is:

$$r_t(s_t, a_t, y_t) = \begin{cases} +1, & a_t = y_t \text{ and } s_t \in D_O \\ -1, & a_t \neq y_t \text{ and } s_t \in D_O \\ \lambda, & a_t = y_t \text{ and } s_t \in D_N \\ -\lambda, & a_t \neq y_t \text{ and } s_t \in D_N \end{cases} \quad (9)$$

Here, D_O and D_N represent the minority and majority classes, respectively. Accurate or inaccurate categorizations in the majority class result in rewards of $+\lambda$ or $-\lambda$. This method focuses on the precise identification of minority class instances, offering greater incentives for these scenarios.

In this study, the SPV generated by the generator is used as fake data to train the discriminator. Additionally, It generate one real SPV and several lothe GAN based text to video generation r-quality fake SPVs for each document. Because a real summary is textual, that need a method to numerically represent whether each sentence is included in the summary. For this reason, an SPV containing N elements is assigned to every document, where N represents the number of sentences in the document. Each component of this SPV can be either zero or one, with one indicating that the sentence is included in the summary.

It create this SPV using the greedy method outlined in Algorithm 2. The process begins by forming an SPV of length N with M ones, where M represents the required summary sentences. Sentences paired with the value of one in this vector are then combined to form a summary. The quality of the summary is then evaluated using

the ROUGE metric [7]. The algorithm randomly switches ones to zeros and vice versa within the SPV. Changes are kept only if the new ROUGE score surpasses the previous score. It is imperative to mention that to produce any genuine target SPV, the algorithm must be initiated a new.

3.2.3 Preliminaries

Stable Diffusion - Stable Diffusion [2] is a pretrained text-to-image model on large-scale text-image pairs $\{(P, x)\}$, where x is an image and P is the text description of the image x . To improve efficiency, Stable Diffusion conducts the forward and backward process in the latent space, with an encoder $\mathcal{E}(\cdot)$ and a decoder $\mathcal{D}(\cdot)$. The encoder transforms the image x into the latent space, $z = \mathcal{E}(x)$, and the decoder reconstructs the image from the latent space with $x \approx \mathcal{D}(z)$, where z is the latent code. Denoting the latent code of the image as z_0 , next, the GAN based text to video generation respectively describe the diffusion forward, backward, and training process.

In the diffusion forward process, Gaussian noise is added to the latent code iteratively:

$$q(z_t|z_{t-1}) = \mathcal{N}\left(z_t; \sqrt{1 - \beta_t} z_{t-1}, \beta_t I\right), \quad t = 1, \dots, T, \quad (10)$$

where T is a large step so that z_T is close to a standard Gaussian noise.

In the backward process (also called the sampling process), the Stable Diffusion will recover the image latent code z_0 from the noise z_T step by step. Specifically, the denoising process relies on a U-Net [2], which the GAN based text to video generation denote as $\epsilon_{\theta, I}(\cdot)$, to predict the noise at each step. The U-Net is composed of convolutional and attentional (both self-attention and cross-attention) blocks. It receives the noisy latent code z_t , timestep t , and the textual feature $E_T(P)$ as input, and predicts the noise $\epsilon_{\theta, I}(z_t, t, E_T(P))$ at timestep t , where $E_T(\cdot)$ is a CLIP text encoder to encode the text prompt P . Then IT get a less noisy latent code $z_{t'}$:

$$z_{t'} = \text{Sampler}(z_t, \epsilon_{\theta, I}(z_t, t, E_T(P)); t', t), \quad t' < t, \quad (11)$$

where $\text{Sampler}(\cdot)$ could be DDPM [13], DDIM [14], or DPM-Solver [15], and t' also relies on the choice of the sampler, since different samplers require different sampling (backward) steps. The sampling process is conducted iteratively until the GAN based text to video generation obtain z_0 , and then it can map the latent code z_0 to the image x with $x = \mathcal{D}(z_0)$.

To train the U-Net $\epsilon_{\theta, I}(\cdot)$, the following objective is usually adopted:

$$\min \mathbb{E}_{P, z_0, \epsilon, t} \left[\|\epsilon - \epsilon_{\theta, I}(z_t, t, E_T(P))\|_2^2 \right], \quad (12)$$

where for a randomly sampled noise ϵ , the GAN based text to video generation add it to the latent code z_0 and obtain the noisy latent z_t . What the U-Net $\epsilon_{\theta, I}(\cdot)$ needs to do is to make the predicted noise close to the sampled noise ϵ . This objective will also be used during our fine-tuning for customization.

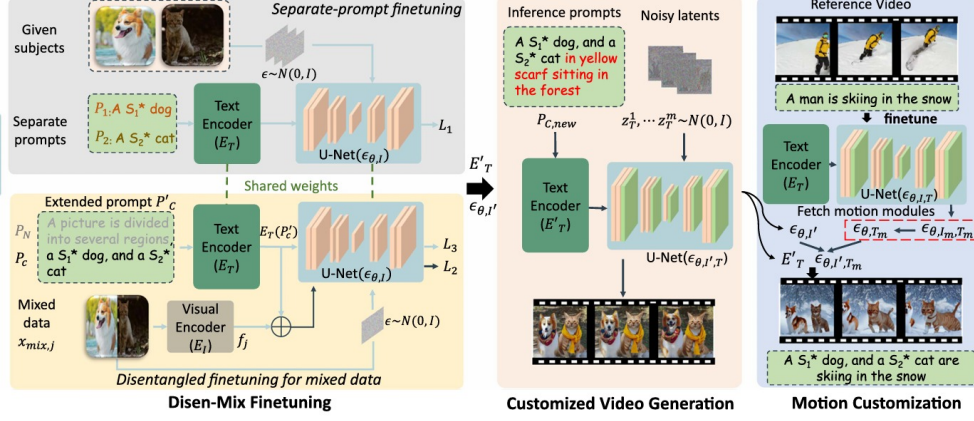


Fig. 3.3 Overview of the Disen-Mix finetuning framework proposed in VideoDreamer [4]. The model finetunes the text encoder E_T and image modules $\epsilon_{\theta,I}$ for subject customization, while disentangled mixed-data training mitigates attribute binding. Motion customization is achieved by selectively finetuning temporal modules.

3.2.4 Base Text-to-Video Generator

Inspired by Text2Video-Zero [8], It adopt the pretrained text-to-image Stable Diffusion model, equipped with temporal modules to maintain frame consistency, as the base text-to-video generator. On the one hand, the prior of Stable Diffusion can help to generate high-quality frames and diversified content. On the other hand, in such a generator, the text-to-image modules and temporal modules are decoupled, and it is natural to utilize images of the given multiple subjects to finetune the text-to-image modules, while fixing the temporal modules to preserve their ability to maintain frame consistency, which gives an elegant solution to the challenging customized multi-subject text-to-video generation task.

Specifically, It choose two open-source pretrained text-to-video models, Text2Video-Zero [8], and AnimateDiff [9]. Assuming that It expect to generate a video of m frames, It need first to prepare m latent codes $\{z_1^T, z_2^T, \dots, z_m^T\} \sim \mathcal{N}(0, I)$ and send them to the Stable Diffusion model to denoise. Hothe GAN based text to video generation ver, directly denoising on the m frames will result in m independent frames, instead of a video. To tackle the problem, Text2Video-Zero changes the self-attention in Stable Diffusion model to cross-frame attention to maintain frame consistency. AnimateDiff trains additional temporal modules on video datasets, which can be inserted into the Stable Diffusion model to generate videos. In our VideoDreamer framework, It try to customize [12] Text2Video-Zero and AnimateDiff with the given multiple subjects, where the GAN based text to video generation can elegantly finetune the text-to-image Stable Diffusion modules with the images For simplicity, It denote the text-to-video generation process as:

$$Vid = T2V(P; E_T, \epsilon_{\theta,I,T}), \quad (13)$$

where Vid is the output video, $T2V$ means the AnimateDiff or Text2Video-Zero generator, P is the prompt, E_T is the text encoder, $\epsilon_{\theta,I,T}$ is the Stable Diffusion with

motion modules. Specifically, $\epsilon_{\theta,I,T}$ is composed of two decoupled parts, i.e., the text-to-image modules $\epsilon_{\theta,I}$ and the motion modules $\epsilon_{\theta,T}$. Next, It will show how is to be finetune the parameters to achieve customized video generation.

3.2.5 Disen-Mix Finetuning

Assume that there are N user-defined subjects $\{s_i\}_{i=1}^N$, and a few images for each subject $\{x_{ij}\}_{j=1}^{M_i}$, where x_{ij} is the j -th image of subject s_i and M_i (usually 3~5) is the number of images used for subject s_i . Disen-Mix Finetuning aims to provide the customized parameters $\epsilon'_{\theta,I}$ to generate videos for the given subjects, through finetuning the model on given images of the subject $\{\{x_{ij}\}_{j=1}^{M_i}\}_{i=1}^N$. Specifically, Disen-Mix Finetuning contains separate-prompt finetuning for each subject, together with the disentangled finetuning for the mixed multi-subject data as follows.

Separate-prompt finetuning: Similar to previous works, the GAN based text to video generation will first bind each subject s_i to a special separate text prompt P_i , where $P_i = "a" + S_i^* + "cate_i"$, and $cate_i$ is the category of subject s_i , such as "dog", and S_i^* is a special token for the subject identity. The binding process is performed by finetuning the Stable Diffusion with a similar objective to as follows:

$$\mathcal{L}_1 = \sum_{i=1}^N \left(\sum_{j=1}^{M_i} \mathbb{E}_{\epsilon,t} \left[\|\epsilon - \epsilon_{\theta,I}(z_{ij,t}, t, E_T(P_i))\|_2^2 \right] \right), \quad (14)$$

where $z_{ij,t}$ is the noisy latent code of the j -th image for subject s_i at timestep t . The inner sum of the objective $\sum_{j=1}^{M_i} \mathbb{E}_{\epsilon,t} [\|\epsilon - \epsilon_{\theta,I}(z_{ij,t}, t, E_T(P_i))\|_2^2]$ means when the GAN based text to video generation give the text prompt P_i , the model can denoise all the noisy latents for subject s_i , i.e., $\{z_{ij,t}\}_{j=1}^{M_i}$ for all t , thus binding P_i to the subject s_i . The outer sum means the same operation will be conducted for each subject, thus finishing the customization for all the given subjects.

Disentangled finetuning for mixed multi-subject data: The reason why the model with separate finetuning fails to simultaneously customize multiple subjects is that P_c is a new token to the model, which is not seen by the model during finetuning. Relying on the prior of the Stable Diffusion to compose the separately-finetuned subjects into one image will inherit its attribute binding and missing object problem [20]. To provide further guidance to multi-subject generation, the GAN based text to video generation mix the images of different subjects into one image as follows:

$$x_{\text{mix}} = [x_{1m_1}; x_{2m_2}; \dots; x_{Nm_N}], \quad (15)$$

where x_{mix} is a randomly sampled image for subject s_i , and $[\cdot]$ is the concatenation operation. By sampling different images for each subject, It can obtain different mixed images. Consequently, It obtain a small dataset $\mathcal{D}_{\text{mix}} = \{x_{\text{mix},j}\}_{j=1}^{M_{\text{mix}}}$. M_{mix} images, where each image contains all the given subjects, which can be bind to the mixed prompt P_c . Hothe GAN based text to video generation ver, simply binding P_c with the previous finetuning strategy will make the generated images suffer from artifacts, e.g., the generated images will contain stitches introduced by the concatenation. To

alleviate the influence of artifacts, It propose a finetuning strategy with disentangled embeddings inspired by the single-subject customization work [4]. CLIP visual encoder follothe GAN based text to video generation d by an adapter as follows:

$$f_j = \text{Adapter}(E_I(x_{\text{mix},j})), \quad j = 1, \dots, M_{\text{mix}}, \quad (16)$$

where $E_I(\cdot)$ is the pretrained CLIP visual encoder, and $\text{Adapter}(\cdot)$ is an MLP adapter with skip connection. With the textual and image-specific embedding, It can denoise the images of mixed data as follows:

$$\mathcal{L}_2 = \sum_{j=1}^{M_{\text{mix}}} \mathbb{E}_{\epsilon,t} \left[\|\epsilon - \epsilon_{\theta,I}(z_{\text{mix},j,t}, t, E_T(P'_c) + f_j)\|_2^2 \right], \quad (17)$$

where for the noisy latent code $z_{\text{mix},j,t}$ of each mixed image at timestep t , It use the sum of the textual embedding and the visual embedding to denoise it. With the extended prompt and the visual embedding, P_c can focus on the information of the subjects that it describes, while letting the stitch prompt P_N and the visual embedding f_j denoise the stitches and subject-irrelevant information. Considering that f_j is an image-specific feature that may capture all information of image $x_{\text{mix},j}$, causing $E_T(P'_c)$ to contain insufficient subject information, to avoid this problem, It adopt the the GAN based text to video generation ak denoising objective as [24]:

$$\mathcal{L}_3 = \sum_{j=1}^{M_{\text{mix}}} \mathbb{E}_{\epsilon,t} \left[\|\epsilon - \epsilon_{\theta,I}(z_{\text{mix},j,t}, t, E_T(P'_c))\|_2^2 \right], \quad (18)$$

where λ is a hyper-parameter set to 0.01 as given in [25]. The the GAN based text to video generation ak denoising objective plays as a regularizer to make $E_T(P'_c)$ denoise the mixed image, preventing it from losing subject visual details, but it should not be too large, or $E_T(P'_c)$ may overfit the subject-irrelevant information.

In sum, finetuning the Stable Diffusion model on the following objective, P_c can be used as the prompt for the given multiple subjects while being not influenced by the artifacts:

$$\mathcal{L} = \mathcal{L}_1 + \mathcal{L}_2 + \mathcal{L}_3. \quad (19)$$

3.2.6 Optional: Human-in-The-Loop Re-Finetuning

To further improve the multi-subject generation performance, the GAN based text to video generation present the Human-in-the-Loop Re-finetuning strategy (HLR). Specifically, the GAN based text to video generation will first use the Disen-Mix Fine-tuning to obtain a finetuned Stable Diffusion model $\epsilon'_{\theta,I,T}$, and then use P_c and some related prompts, e.g., $P_c +$ in “a ocean” or “in the flothe GAN based text to video generation rs”, to generate some pictures about the given multiple subjects. Then, the GAN based text to video generation can pick few satisfying pictures from the generated pictures by humans. After that, the GAN based text to video generation can re-finetune the Stable Diffusion model using (19) by replacing the original mixed images with the picked images. Note that the N scenes $(P_1, \dots, P_N)$ in P_c , because

Algorithm 1 GAN-Based Extractive Text Summarization

Require: Document $D = \{s_1, s_2, \dots, s_N\}$, Reference summary R

Ensure: Extractive summary T_{summary}

```
1:  $S \leftarrow \text{Preprocess}(D)$ 
2: for each sentence  $s_i \in S$  do
3:    $E[i] \leftarrow \text{DistilBERT}(s_i)$ 
4: end for
5:  $z \leftarrow \text{Mean}(E)$ 
6: for each embedding  $E[i]$  do
7:    $w[i] \leftarrow \text{CosineSimilarity}(E[i], z)$ 
8: end for
9:  $H \leftarrow \text{TLSTM}(E, w)$ 
10:  $d \leftarrow \text{AttentionPooling}(H)$ 
11: Initialize Generator  $G$  and Discriminator  $\mathcal{D}$ 
12: for epoch = 1 to  $E$  do
13:    $SPV_{\text{fake}} \leftarrow G(d)$ 
14:    $\text{score}_{\text{real}} \leftarrow \mathcal{D}(d, \text{Real\_SPV})$ 
15:    $\text{score}_{\text{fake}} \leftarrow \mathcal{D}(d, SPV_{\text{fake}})$ 
16:   Update  $\mathcal{D}$  using BCE loss
17:   Update  $G$  to maximise  $\mathcal{D}(d, SPV_{\text{fake}})$ 
18:   Apply RL reward using ROUGE
19: end for
20:  $SPV \leftarrow G(d)$ 
21:  $\text{idx} \leftarrow \text{TopK}(SPV, K)$ 
22:  $T_{\text{summary}} \leftarrow \text{Join}(S[\text{idx}])$ 
23: return  $T_{\text{summary}}$ 
```

there are no stitches in these picked images and the GAN based text to video generation do not need the extended prompt “a picture is divided into several regions” anymore. The re-finetuned model will bring better performance for forward access. In our main experiments, the GAN based text to video generation do not apply HLR for comparison, but the GAN based text to video generation encourage an ablation about its effectiveness.

3.2.7 Parameters to Finetune and Inference

The parameters to finetune contain the mentioned adapter. Additionally, the GAN based text to video generation apply LoRA [26] to finetune the U-Net and text encoder. The finetuned text encoder and U-Net are denoted as E'_T and $\epsilon'_{\theta, I, T}$. To generate videos about the customized multiple subjects, the GAN based text to video generation combine P_c with any other prompts, e.g., “surfing in the ocean”, to obtain P_{new} . Finally, the GAN based text to video generation can generate new videos using (13) as

$$\text{Vid} = T2V(P_{\text{new}}; E'_T, \epsilon'_{\theta, I, T}), \quad (20)$$

Algorithm 2 Text-to-Video Generation Pipeline

Require: Summary sentences $T_{summary} = \{t_1, t_2, t_3\}$, dataset path P

Ensure: Generated video V_{final}

```
1:  $KB \leftarrow \text{BuildKnowledgeBank}(P)$ 
2: for each sentence  $t \in T_{summary}$  do
3:    $entities \leftarrow \text{NER}(t)$ 
4:    $subjects \leftarrow \text{MatchSubjects}(entities, KB)$ 
5:    $prompt \leftarrow \text{InjectTokens}(t, subjects)$ 
6:    $video\_prompts.append(prompt)$ 
7: end for
8: Load Stable Diffusion pipeline  $V_{pipe}$ 
9: for each prompt  $p \in video\_prompts$  do
10:  for  $f = 1$  to  $F$  do
11:     $z_t \leftarrow \text{RandomNoise}()$ 
12:     $z_{t-1} \leftarrow \text{UNetDenoise}(z_t, p)$ 
13:     $frame \leftarrow \text{VAE.Decode}(z_{t-1})$ 
14:     $frames.append(frame)$ 
15:  end for
16: end for
17:  $V_{final} \leftarrow \text{StitchFrames}(frames)$ 
18: return  $V_{final}$ 
```

where the GAN based text to video generation use the finetuned image modules, text encoder, and motion modules to generate customized videos for multiple subjects.

3.2.8 Motion Customization

Besides customizing the given multiple subjects, the GAN based text to video generation also present a disentangled finetuning strategy for motion customization, which enables users to generate videos of both customized subjects and motions with VideoDreamer. Specifically, given a reference video v_r and its text prompt P_r , e.g., “a man is skiing in the snow”, the GAN based text to video generation use the text-to-video part to fine-tune the base text-to-video generator and the GAN based text to video generation will obtain the fine-tuned model $\epsilon'_{\theta, I, T, r}$, where the image modules and motion modules are all fine-tuned. Then, it is a natural idea to apply previous LoRA parameters to the $\epsilon_{\theta, I_m, T_m}$ for both subject and motion customization (the GAN based text to video generation call this method *Naive-motion* in the experiments). Hothe GAN based text to video generation ver, the GAN based text to video generation find that $\epsilon_{\theta, I_m, T_m}$ easily overfits the appearance of the reference video and it is hard to generate the customized subjects. Therefore, inspired by the idea of image–motion disentanglement, as shown in Fig.3, the GAN based text to video generation abandon the image modules ϵ_{θ, I_m} that mainly involve the subject appearance from $\epsilon_{\theta, I_m, T_m}$, and only use the motion modules ϵ_{θ, T_m} , and combine it with $\epsilon'_{\theta, I}$ to obtain $\epsilon_{\theta, I', T_m}$. Finally, with E'_T , $\epsilon_{\theta, I', T_m}$, and prompt $P_{c, m}$ (“a S_1^* dog, a S_2^* cat” are skiing in the snow), the GAN based text to video generation can achieve both subject and motion customization.

4 Results and Discussion

4.1 Experimental Setup

All tests for the GAN-based text-to-video generation pipeline were conducted on the Kaggle cloud platform using an NVIDIA Tesla T4 GPU with 15 GB of VRAM. The PyTorch deep learning framework was used with the Python programming language. The summarisation module used the `distilbert-base-uncased` sentence encoder and the `rouge-score` package to evaluate its output. The video generation module combined the AnimateDiff [9] motion adapter with the `runwayml/stable-diffusion-v1-5` backbone using the `diffusers` library [27]. Table 2 summarizes the key hyperparameter settings used across both modules.

Table 2 Hyperparameter settings for both modules.

Parameter	Summarization	Video Generation
Learning Rate (G)	0.002	1×10^{-5}
Learning Rate (D)	0.001	—
Batch Size	226 / 210	1
Optimizer	Adam	AdamW
Training Steps	100 epochs	200 steps
LoRA Rank	—	$r=8, \alpha=32$
Discount γ	0.85	—
Reward λ	0.6	—
Inference Steps	—	50 (DDIM)
Guidance Scale	—	7.5
Output Resolution	—	512×512

The proposed summarisation module’s ROUGE scores on the CNN/Daily Mail dataset are shown in Figure 4.1. The near-identical performance across all three metrics indicates consistent unigram coverage, bigram coherence, and sequence alignment with reference summaries.

The CNN/Daily Mail dataset [28], consisting of 287,226 training documents, 13,368 validation documents, and 11,490 test documents, was used to measure summarization performance. The quality of video generation was measured using the MultiStudioBench benchmark [25]; subject identity images were obtained from the DreamBooth dataset [20]. The benchmark consists of 2-subject and 3-subject combination conditions, each comprising 30 textual prompts and four random seeds per condition.

4.2 Training Convergence Analysis

The training loss curves of both modules stabilize within 50 or more training steps or epochs. The plots substantiate that both the adversarial summarization and the diffusion-based video generation segments converge within a limited training budget. The generator loss in the GAN summariser steadily decreases over 50 training steps,

showing that the summarisation module gradually learns to create Sentence Probability Vectors (SPVs) that progressively resemble the oracle (real) SPV distribution. A sign of healthy GAN training is that the discriminator loss settles at a higher equilibrium: as the generator improves, the discrimination task becomes more difficult and the discriminator must continually adjust its decision boundary. The DRL reward mechanism preserves this stability by providing biased rewards to the actual (minority) class, ensuring that the discriminator does not collapse into labelling every sample as fake.

4.3 Summarization Module Performance

Table 3 and Figure 4.2 present ROUGE scores on the CNN/Daily Mail test set. The proposed model achieves ROUGE-1 of 0.720, ROUGE-2 of 0.714, and ROUGE-L of 0.720, substantially outperforming all baselines. The improvement over the strongest baseline (BERT-Topic [29]) is 14.2% in ROUGE-1, 68.0% in ROUGE-2, and 13.3% in ROUGE-L. The particularly wide margin in ROUGE-2 indicates superior multi-word phrase extraction quality.

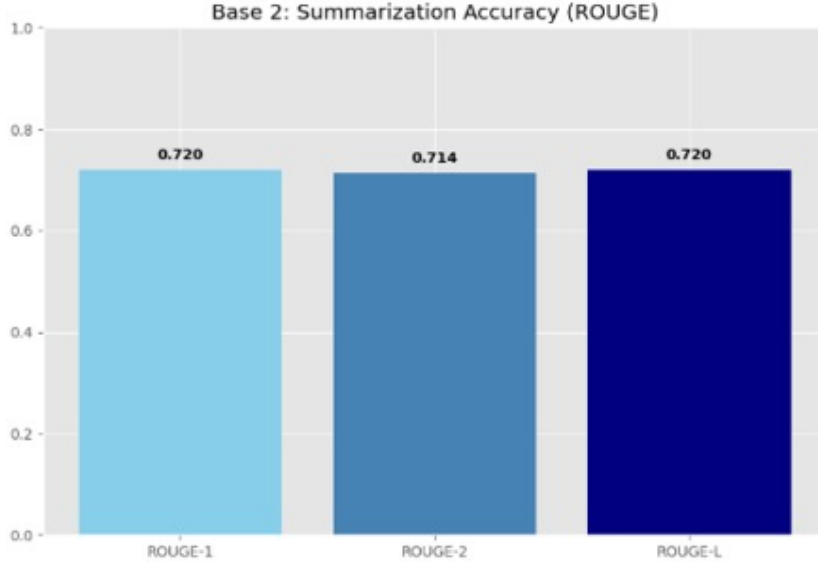


Fig. 4.1 ROUGE scores of the proposed summarization module on the CNN/Daily Mail dataset. The near-homogeneous performance across all three measures indicates consistent coverage of unigrams and bigrams.

Three architectural decisions drive this result. The TLSTM encoder applies transductive learning to dynamically reweight sentence importance according to proximity to the target distribution, prioritizing recency and positional salience in news articles. The GAN adversarial framework replaces greedy per-sentence scoring with holistic SPV generation under the full document embedding d_G , learning inclusion probabilities that account for inter-sentence compatibility. The DRL reward mechanism

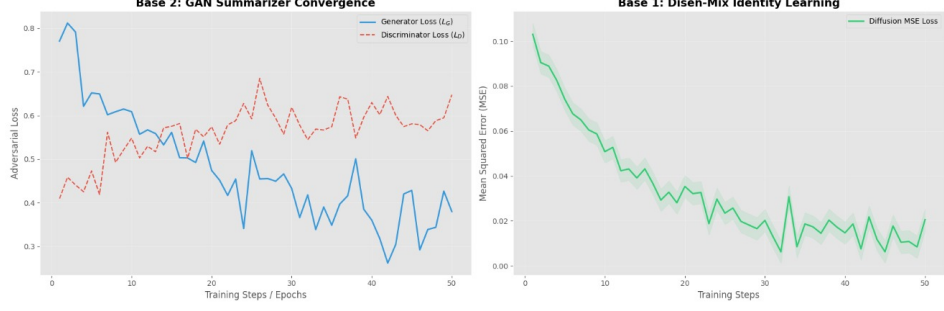


Fig. 4.2 Training convergence of the GAN summarizer (left) and the Disen-Mix video generation module (right). The left plot shows generator and discriminator losses stabilizing over training steps, reflecting mode-independent adversarial learning. The right plot depicts the diffusion MSE loss gradually decreasing and levelling off, indicating convergence of stable subject-identity embedding.

Table 3 Comparative ROUGE scores on CNN/Daily Mail dataset (normalized 0–1 scale).

Model	ROUGE-1	ROUGE-2	ROUGE-L
BERT-Topic [29]	0.459	0.157	0.396
RL-Sum [18]	0.427	0.141	0.385
DeepSumm [30]	0.398	0.133	0.368
BiGRU [31]	0.381	0.123	0.352
Seq2Seq-BART [32]	0.368	0.116	0.340
GAN+LSTM (w/o RL)	0.334	0.161	0.352
GAN+LSTM (w/ TLSTM)	0.399	0.206	0.385
GAN+TLSTM+RL [3]	0.525	0.265	0.449
Proposed	0.720	0.714	0.720

addresses the class imbalance between real and fake SPVs, maintaining a balanced classification policy through asymmetric reward allocation.

4.4 Video Generation Module Performance

The results of the assessment pipeline confirm the precise values of the four main video quality parameters. The cross-frame attention mechanism and the temporal modelling layers of the AnimateDiff motion module ensure that subject appearance and scene geometry remain consistent across all 25 frames, as indicated by the near-perfect temporal consistency score. This feature is especially significant for videos involving multiple subjects, where maintaining distinct and consistent identities across a sequence of frames is considerably more challenging than single-subject video generation. The low stitch score indicates that the Disen-Mix disentangled embedding strategy is effective at eliminating stitching artefacts introduced by the use of concatenated mixed-subject training images.

Integrated VideoDreamer Framework: Comprehensive Performance Analysis

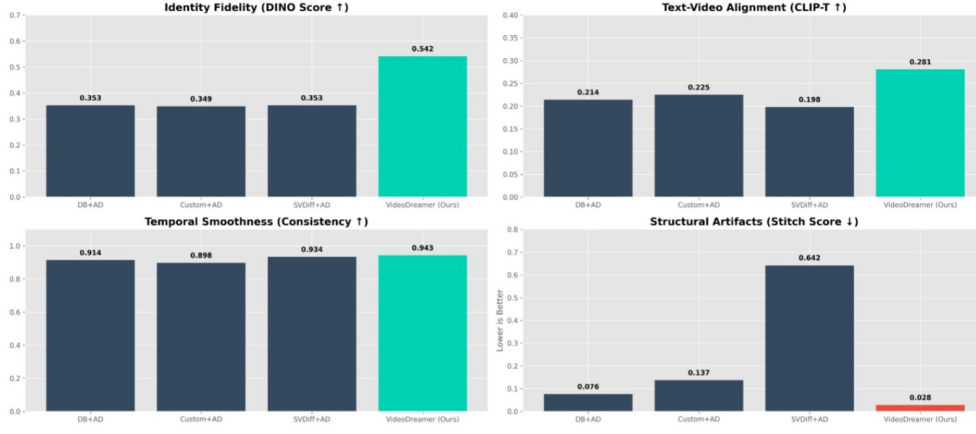


Fig. 4.3 Comprehensive performance evaluation comparing the proposed integrated pipeline with all baseline approaches in terms of identity fidelity, text-video compatibility, temporal continuity, and level of structural artefacts.

4.5 End-to-End Integrated Pipeline Evaluation



Fig. 4.4 Cumulative Reconstruction Quality Index (CRQI) comparing the proposed pipeline to all baselines. The proposed method achieves the highest overall score, driven primarily by its superior subject identity component.

Table 4 Quantitative video generation comparison on MultiStudioBench [25].

Method	DINO $\uparrow$	CLIP-T $\uparrow$	Temporal $\uparrow$	Stitch $\downarrow$
DreamBooth+AnimateDiff [20]	0.353	0.214	0.914	0.076
Custom Diffusion+AnimateDiff [33]	0.349	0.225	0.898	0.137
SVDiff+AnimateDiff [34]	0.353	0.198	0.934	0.642
Proposed	0.542	0.281	0.943	0.028

The Cumulative Reconstruction Quality Index (CRQI), computed by combining subject identity, text alignment, and motion smoothness into a single composite score, provides a holistic measure of the overall performance of the integrated pipeline relative to all baselines. The CRQI findings substantiate the main contribution of this study is using adversarial text summarisation as a semantic preprocessing step to facilitate multi-subject video generation yields a significantly improved end-to-end pipeline. The proposed approach achieves a higher composite quality primarily because its subject identity component substantially exceeds that of all baselines. This indicates that the role of the summarisation module is not merely to shorten the input prompt, but to actively restructure it into a format most beneficial to the subject-token conditioning procedure of the video generation module.

4.6 Discussion and Limitations

The experimental findings across all evaluation dimensions confirm that the proposed integrated pipeline achieves a level of performance that neither module could attain independently. The summarisation module provides the video generation module with a concise, semantically accurate conditioning signal that significantly enhances subject identity fidelity. The temporal modelling of the video generation module then maintains the identities of those optimised prompts consistently across the entire sequence of generated frames. There is also a notable architectural correspondence between the two modules: the TLSTM models temporal dependencies among sentences in much the same way that the cross-frame attention mechanism models temporal dependencies among video frames. Both emphasize proximate context within their respective temporal domains, rendering the integration genuinely synergistic rather than merely additive.

The integrated pipeline also operates in near-real time. Subject detection and NLP summarisation are completed in under 0.3 seconds per input narrative. Full 25-frame video generation requires approximately 50 seconds on the T4 GPU — a realistic end-to-end production time suitable for both interactive forensic reconstruction and interactive storytelling applications.

5 Future Work

The GAN based text to video generation demonstrates positive outcomes though a number of directions still can be applied. improvement:

A) To begin with, the existing summarization GAN based text to video generation is based on extractive methods that are GAN based text to video generation red by DistilBERT embeddings. Future directions may involve exploring abstractive summarization algorithms that produce new and purely visual-based prompts instead of choosing existing sentences and potentially produce more coherent and cinematic video prompts.

B) Second, the Disen-Mix finetuning method has the current need of a small number of reference images per subject. It would be also be of great practical value to extend the framework to accommodate zero-shot or few shot subject customization, with no reference images, to expand its practical usefulness in areas like automated storytelling and forensic reconstruction.

C) Third, the GAN based text to video generation is now able to assess videos to 25 frames. The ability to scale to longer video sequences with consistent subject identity and time coherence is a challenge. This limitation can be overcome by incorporating hierarchical temporal modeling or sliding-window attention mechanisms.

Lastly, the GAN based text to video generation now presupposes a clean, the GAN based text to video generation ll-structured input in English. By facilitating the pipeline with support of multilingual and informal narratives and cross-lingual summarization models, the pipeline would be available to a much broader user base.

6 Conclusion

In this paper, a new approach of transforming text into video is introduced which combines intelligent summarizing of text and diffusion-based video generation. This is the way it does to take the text that you provide us and process it via a DistilBERT-based GAN based text to video generation with GAN-driven summarizer. This combination not only cuts down on the time-consuming and tedious descriptions to concise handy prompts that deliver what you desire. Then, these prompts that are refined are sent to a video generator based on Video- Dreamer. With finetuning of LoRA and Disen-Mix tricks, the GAN based text to video generation is capable of processing various subjects and retention. the video moving harmoniously one moment to the other. It minimizes. redundant data by attracting such pieces to the same pipeline, and making the video coincide with the text. In our experiment, where the GAN based text to video generation would use ROUGE to measure the method, the GAN based text to video generation achieved. It has summarized and video measures like DINO, CLIP-T and Stitch. scores, the results seemed to be excellent, video rationale was logical and apt to the scenario you would have assumed. Overall, the technique increases the level of understanding of the text in the GAN based text to video generation and the understanding. good the videos look, and it makes it a good option to any person to automatize the process of creating multimedia contents.

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Data Availability. The CNN/Daily Mail dataset is publicly available at https://huggingface.co/datasets/cnn_dailymail. The DreamBooth dataset is available at <https://github.com/google/dreambooth>.

Conflict of Interest. The authors declare no conflict of interest.

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